PROPOSAL · COMPUTE-GOVERNANCE INFRASTRUCTURE

Building the Pause Button

The paper proposes technical and institutional infrastructure that could make a future global compute pause enforceable, with thresholds set and revised through governance.

Background and provenance

WHY IT WAS WRITTEN

The authors saw a gap between calls for a global compute pause and the infrastructure that would let governments define, enforce, and verify one. They argue that rapid capability growth creates a shrinking window in which to build credible controls before a dangerous training run occurs. The paper organizes hardware controls, supply-chain traceability, and legal rules into a governance–enforcement–verification framework rather than claiming one device could serve as a complete pause button.

PEOPLE AND ORGANISATIONS

The manuscript lists Ananthi Al Ramiah, Raymond Koopmanschap, Josh Thorsteinson, Sadruddin Khan, Jim Zhou, and Shafira Noh as independent; Joep Meindertsma is affiliated with PauseAI, and Farhan Shafiq with Think Safe AI. Its acknowledgments say the 10th AI Safety Camp convened and supported the research team. These are several affiliations and a project setting, not a single organisation authoring or officially adopting the proposed regime.

KEY DATES

The arXiv record lists one submission on 25 June 2025. The HTML manuscript also displays an internal date of 24 August 2026, although the arXiv record shows no later version. The two dates should be read separately: the former is the recorded public submission date; the latter appears in the manuscript and its provenance is not explained there.

EARLIER FOUNDATIONS

The technical menu builds on earlier compute-governance and hardware-verification research, including 2024 work on flexible hardware-enabled guarantees and offline licensing. The paper also uses nuclear and chemical-weapons verification and export-control regimes as institutional analogies. Its own contribution is to place those components inside a proposed global architecture; a cited technique should not be mistaken for a device the authors had already built.

Sources: Manuscript, affiliations and acknowledgments ↗ arXiv submission record ↗

AUTHORS	Ananthi Al Ramiah, Raymond Koopmanschap, Josh Thorsteinson, Sadruddin Khan, Jim Zhou, Shafira Noh, Joep Meindertsma, and Farhan Shafiq
ORGANISATION / AFFILIATION	Independent authors; PauseAI; Think Safe AI
ORIGINAL DOCUMENT	**Toward a Global Regime for Compute Governance: Building the Pause Button ↗**

PURPOSE

It focuses on catastrophic AI risks, including loss of control and harmful misuse, while also discussing societal disruption and concentrated power. Its near-term objective is to create credible capacity to slow dangerous frontier development before a crisis.

WHAT IS PACED

Training runs or effective cluster capacity above agreed limits. The paper uses example thresholds, including a 10^{25}-FLOP training ceiling, as design illustrations rather than a settled treaty number. Chip supply, cluster interconnects, and model access are supporting control points.

WHEN AND FOR HOW LONG

Build the governance and verification stack now, before a dangerous frontier capability makes an emergency pause necessary. It does not commit to a specific date for activating a global halt or a fixed duration for one.

HOW PACING WORKS

Combine compute caps and licensing with hardware measures such as operation counters or bandwidth limits, supply-chain traceability, model locking, export controls, and inspection rights. The paper frames this as governance, enforcement, and verification layers.

VERIFICATION METHODS

Chip and component registries, chain-of-custody records, facility and cable inspections, hardware FLOP counters or secure receipts, and identity-linked cloud-use records help check declarations. Capability evaluations may inform threshold revisions but do not themselves prove compute compliance.

WHO RUNS IT

National regulators would license and enforce controls, coordinated through a global regime with independent auditing and inspection capacity. The paper draws on international verification institutions as analogies without settling the precise legal body.

LIMITATIONS

The architecture requires broad political cooperation, reliable hardware and supply-chain engineering, and protection against tampering. Compute is an imperfect risk proxy as algorithms improve, so thresholds and safeguards would need revision.

Mapped verification methods

10 mechanisms in the atlas map to this text. The labels indicate the strength of each match, not whether the method works.

Chip production and transfer ledger
Core

Cloud customer identity and account linkage
Core

Interconnect bandwidth cap
Core

On-chip operation metering
Core

Cloud workload metering
Specified option

Facility portal and camera monitoring
Specified option

Physical chip inventory inspection
Specified option

Signed hardware or scheduler audit logs
Specified option

Independent predeployment capability evaluation
Partial match

Weight custody and egress monitoring
Partial match

Explore the comparison

See all eight proposals by mechanism →

This is an analytical summary of the linked proposal. Examples and scenario dates are identified as such; see the original document for the authors' full arguments.